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CNAS L4743

检测报告

报告号: AJFS2005003594RS_CN

日期: 2020 年 5 月 26 日 第 1 页 共 7 页

波士胶(上海)管理有限公司

上海市闵行区光华路 968 号 1 号楼

以下测试样品为客户提供并确认, SGS 不对这些数据 and 信息的真实性、完整性以及由此引发的结果和/或结论的有效性负责。检测结果适用于收到的样品。

样品名称: ISR 7003

颜色: 黑色

型号/货号: /

测试要求:

EN 45545-2:2013+A1:2015 铁路应用—轨道车辆消防保护 第 2 部分: 材料及部件的防火性能要求,
测试参照表格 5—材料要求 (R22)、(R23)

测试结果: -- 参见数据附表--

测试持续时间

样品收到日期 : 2020 年 5 月 14 日

测试执行时间 : 2020 年 5 月 14 日至 2020 年 5 月 25 日

通标标准技术服务有限公司安吉分公司

授权代表签署

邹石龙
实验室经理

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I. 样品信息

样品描述	铝板+胶+铝板
颜色	铝板: 银色; 胶: 黑色
暴露 (测试) 面	任意面
样品尺寸	T01 EN ISO 4589-2: 100mm×10.2mm×5.0mm
	T10.03 EN ISO 5659-2: 75mm×75mm×5.0mm
	T12 NF X70-100-1&-2: 1, <u>1.0020g</u> 2, <u>0.9995g</u> 3, <u>1.0000g</u>
基材	铝板, 1.2mm, 银色

II. 结果概要

要求简称(适用部分)	参考测试方法	参数单位	测试结果 *
R22 (IN16; EL2; EL6A; EL7A; M2)	T01 EN ISO 4589-2: OI	氧指数 %	48.3%
	T10.03 EN ISO 5659-2: 25 kW/m ²	Ds max. 无量纲	1.9
	T12 NF X 70-100-1 及 -2 600℃	CIT _{NLP} 无量纲	0.11
R23 (EX12; EL2; EL5 EL6B; EL7B; M3)	T01 EN ISO 4589-2: OI	氧指数 %	48.3%
	T10.03 EN ISO 5659-2: 25 kW/m ²	Ds max. 无量纲	1.9
	T12 NF X 70-100-1 及 -2 600℃	CIT _{NLP} 无量纲	0.11

* 测试详细信息, 参见本报告的附录。

III. 结论

根据测试结果, 送检样品符合R22&R23 (详见EN 45545-2:2013+A1:2015表格5) 中有关HL3危险等级。

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测试要求 EN 45545-2:2013+A1:2015 表 5 材料综合要求 (R22)、(R23)

要求简称 (适用部分)	参考测试方法	参数单位	定义要求	HL1	HL2	HL3
R22 (IN16; EL2; EL6A; EL7A; M2)	T01 EN ISO 4589-2: OI	氧指数 %	最小	28	28	32
	T10.03 EN ISO 5659-2: 25 kW/m ²	Ds max. 无量纲	最大	600	300	150
	T12 NF X 70-100-1 及 -2 600°C	CIT _{NLP} 无量纲	最大	1.2	0.9	0.75
R23 (EX12; EL2; EL5 EL6B; EL7B; M3)	T01 EN ISO 4589-2: OI	氧指数 %	最小	28	28	32
	T10.03 EN ISO 5659-2: 25 kW/m ²	Ds max. 无量纲	最大	--	600	300
	T12 NF X 70-100-1 及 -2 600°C	CIT _{NLP} 无量纲	最大	--	1.8	1.5

陈述:

此符合性声明仅基于本次实验室活动的实际值, 未将本次实验室活动的结果不确定度影响计入。

本试验结果得出的产品性能是在特殊条件下的检验结果。在实际应用中, 它们不能单独作为评价该产品潜在火灾危险性的依据。

本测试结果仅代表被测试试样。

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附录 1: T01 EN ISO 4589-2:2017 氧指数法测定燃烧特性 第 2 部分:室温氧指数

1. 预处理

T: 23±2°C, R.H: 50±5%条件下, 至少处理 88h。

2. 测试结果

a) 选起始氧浓度(依照 8.2.3): 30%

b) 初始氧浓度的确定: 直到得到两个氧浓度相差≤1 %, 其中一个反应符号为“O”, 另一个反应符号为“X”, 则反应符号为“O”的氧浓度就是初始氧浓度 (根据8.6)

氧浓度, % (V/V)	30	35	40	45	50	49	48	
燃烧长度, (毫米)	<50	<50	<50	<50	>50	>50	<50	
反应, (“X” 或 “O”)	O	O	O	O	X	X	O	

初始氧浓度= 48.0 % (此氧浓度作为第2部分测试的第一个浓度值)

c) 确定极限氧指数 (根据 8.7)

确定氧浓度增加或减少的步长为 $d = 0.2$ % [除非特殊指出, 一般用0.2 % (V/V)]

参数	NT 系列测试										
	NL 系列测试 (8.7.1 和 8.7.2)						(8.7.3)				cf
氧气浓度, % (V/V)	48.0	48.2	48.4	--	--	--	48.4	48.2	48.4	48.2	48.4
燃烧长度, (毫米)	<50	<50	>50	--	--	--	>50	<50	>50	<50	>50
反应, (“X” 或 “O”)	O	O	X	→	→	→	X	O	X	O	X
	k 所在列 (2, 3, 4 或 5): 3						k 所在排 (1 到 16): 6				
	EN ISO 4589-2表4中查得k值为: -0.46										
	故 k= -0.46										

$$OI = Cf + kd = 48.4 + (-0.46 \times 0.2)$$

$$= 48.3\% \text{ (取一位小数)}$$

$$= 48.31\% \text{ (取二位小数)}$$

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附录 2: T10.03 EN ISO 5659-2:2017 塑料—烟生成—第 2 部分:单室法测定烟密度试验方法, 辐射照度 25kW/m² 带有焰模式, 测试时间 10min.

1. 预处理

T: 23±2°C, R.H: 50±5% 条件下, 样品处理至重量恒定状态。

2. 测试结果

参数	1	2	3	平均值
重量 (g)	56.6	59.1	57.6	57.8
D _s (1.5)	0.6	0.5	0.6	0.6
D _s (4)	1.0	1.2	1.1	1.1
D _s (10)	1.4	1.9	2.3	1.9
VOF ₄ min	2.7	3.1	2.9	2.9
D _s max	1.4	1.9	2.3	1.9
T (D _s max) s	600	600	600	600

注释:

D_s (n) —试验进行第n分钟时的比光密度;

VOF₄ —试验前4分钟的比光密度累加值;

D_s max —试验期间最大比光密度。

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附录 3: T12 NF X70-100-1:2006 燃烧试验 烟气分析 第 1 部分:热解法产生气体的分析方法 & NF
X70-100-2:2006 燃烧试验 烟气分析 第 2 部分:管式炉热解法, 炉温设置:600℃, 非列表产品毒性试验.

1. 预处理

T: 23±2°C, R.H: 50±5 %条件下, 至少处理 48h 且样品直至重量恒定状态。

2. 测试结果

气体浓度 [mg/g]	1	2	3	平均值	参考浓度 [mg/m³]
CO	126.75	133.18	135.61	131.85	1380
CO ₂	731.25	786.96	748.66	755.63	72000
HF	ND	ND	ND	--	25
HCl	ND	ND	ND	--	75
HBr	ND	ND	ND	--	99
HCN	ND	ND	ND	--	55
NO, NO _x	ND	ND	ND	--	38
SO ₂	ND	ND	ND	--	262

注: ND---值未发现

计算CIT_{NLP}

$$CIT_{NLP} = 1 \frac{g}{m^3} \sum_{i=1}^{i=8} \frac{Y_i}{C_i}$$

Y_i : NF X70-100-1 管式炉内产生 ith类气体的浓度, 单位 mg/g;

C_i : 第 ith类气体的参考浓度, 单位 mg/m³.

CIT_{NLP} = 0.11

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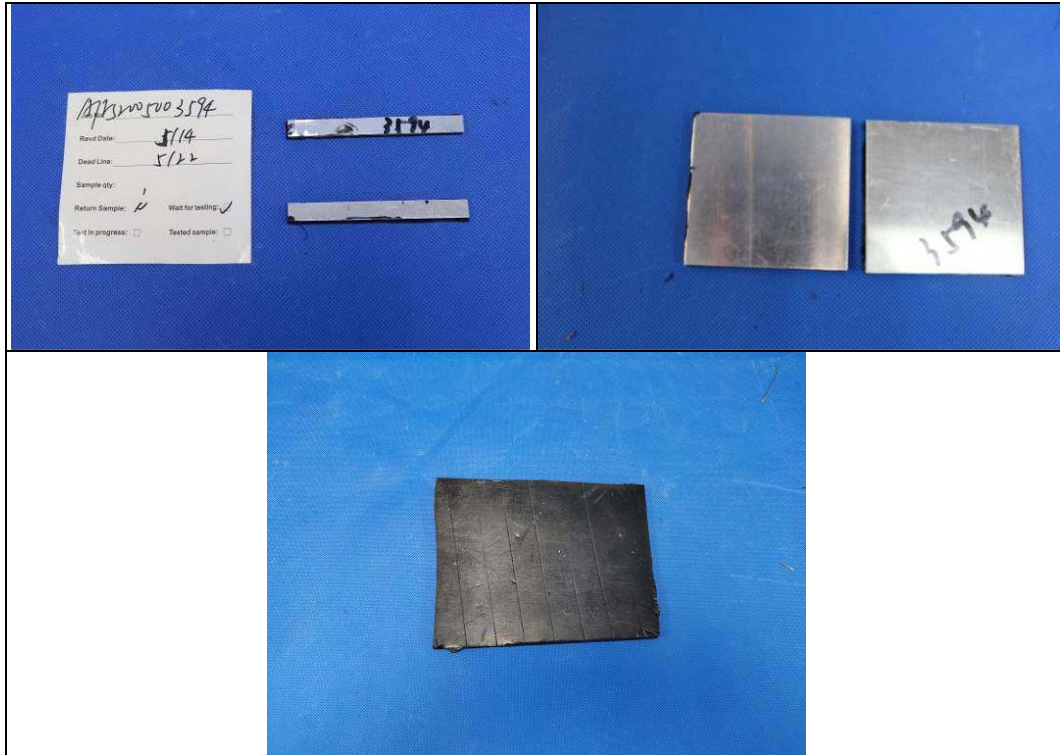
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